

HARICHANDANA MUKKAMALA

DATA ANALYST

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SUMMARY

- **3 years of hands-on experience in data analysis and business intelligence**, using Python (Pandas, NumPy, Scikit-learn) and SQL to deliver actionable insights and support data-driven decisions.
- **Developed interactive dashboards and visual reports** in Tableau, Power BI, and Matplotlib to visualize key metrics and business trends effectively.
- **Applied statistical methods and machine learning techniques** (e.g., regression, A/B testing, trend analysis) to solve business problems and enhance strategic outcomes.
- **Worked with large datasets across cloud platforms** such as AWS and Google Cloud, leveraging relational (MySQL, PostgreSQL) and NoSQL (MongoDB) databases for scalable analysis.
- **Skilled in data storytelling and stakeholder communication**, translating complex data into clear narratives to drive performance and operational improvements.

SKILLS

Methodologies:	SDLC, Agile, Waterfall
Programming Language:	Python, SQL, R, NoSQL, DAX, MATLAB
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn, ggplot2
Visualization Tools:	Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP)
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook
Database:	MySQL, PostgreSQL, MongoDB, SQL Server
Cloud Platform:	Amazon Web Services (AWS), Microsoft Azure
Other Technical Skills:	SSIS, SSRS, Machine Learning Algorithms, ANOVA, ETL Tools, EDA, Ad hoc Analysis, Snowflake, Alteryx, Advance Analytics, SAS, MS Visio, Hypothesis Testing, Regression Analysis, Data Mining, Data Visualization, Data warehousing, Data transformation, Big Data, Data Integration, Clustering, Classification, Regression, A/B Testing, Forecasting & Modelling, Data Cleaning, Data Wrangling, GitHub, JIRA
Soft Skills:	Effective Communication, Problem-Solving, Time Management, Stakeholder Management, Presentation Skills, Risk Management, Flexibility, Networking, Optimization Techniques, Verbal Communication, Analytical Thinking, verbal, and written communication.

EXPERIENCED

Data Analyst, International Bank of Commerce USA, TX	Aug 2024 – Present
▪ Extracted historical stock price data using Yahoo Finance API, integrating it into MySQL databases for structured analysis, reducing data retrieval times by 30% for faster decision-making. ▪ Designed and automated ETL workflows using Python, transforming raw data (stock prices, trading volumes) and loading it into AWS Redshift, improving data processing efficiency by 40%. ▪ Conducted Exploratory Data Analysis (EDA) with Pandas and Seaborn, cleaning and preprocessing data, identifying trends, and reducing data anomalies by 25%, leading to more accurate models. ▪ Developed Long Short-Term Memory (LSTM) models using TensorFlow and Keras, capturing complex temporal patterns and boosting prediction accuracy by 45% over baseline ARIMA models. ▪ Evaluated model performance using RMSE, MAE, and MAPE metrics, refining the predictive model and achieving 40% improvement in forecast reliability compared to previous approaches. ▪ Implemented ANOVA to determine the statistical significance of different features (e.g., technical indicators, market sentiment) on stock prices, increasing the model's explanatory power by 30%. ▪ Conducted A/B testing to evaluate different trading strategies, demonstrating that a combined strategy (e.g., moving averages with volatility indicators) outperformed individual strategies by 20% in backtesting. ▪ Designed interactive dashboards using Power BI, increasing stakeholder engagement by 50% by providing real-time access to predictive insights, portfolio performance, and risk metrics.	

- Wrote complex SQL queries in **MySQL** to aggregate and filter large datasets, reducing the time required for data extraction by 30% and enabling faster reporting for ad-hoc analysis.
- Integrated **Flask with TensorFlow to deploy LSTM-based models as RESTful APIs**, reducing prediction latency by 40%, enabling real-time stock price predictions for business users.
- Managed historical and real-time data storage in AWS S3, streamlining data access and reducing storage costs by 25% through efficient data compression and lifecycle policies.
- Monitored model performance with **AWS CloudWatch**, retraining models with updated financial data to maintain a 20% increase in forecast accuracy, ensuring timely predictions even in volatile markets.

Data Analyst, NextGen Healthcare| India

May 2021 – Jul 2023

- Built predictive models using **Python and Scikit-learn** to forecast ICU bed demand and emergency patient admissions, resulting in a 25% improvement in the hospital's ability to plan resources and reduce overcrowding.
- Designed and deployed real-time **Tableau** dashboards for visualizing hospital resource utilization, including bed occupancy, ventilator usage, and staff availability, increasing executive-level data accessibility and decision-making efficiency by 30%.
- Developed optimized **SQL** queries and ETL workflows to extract, transform, and load complex operational data from hospital information systems, reducing data preparation time by 20% and enabling faster reporting cycles.
- Leveraged **Azure Cloud services (Azure Data Lake, Azure Synapse Analytics)** to store, process, and analyze large-scale healthcare data, improving data storage scalability and enabling efficient analytics workflows across the hospital network.
- Leveraged **Apache Spark** for distributed processing of large-scale hospital operations and EHR datasets, reducing batch data processing time by 35% and improving analytics scalability across multiple departments.
- Utilized **Excel (Pivot Tables, VLOOKUP, advanced formulas)** to automate and validate daily reconciliation reports for medical equipment, patient beds, and supply inventory, increasing data accuracy by 40%.
- Implemented automated workflows in **Alteryx** to streamline the generation of daily and weekly hospital operations reports, reducing turnaround time by 30% and allowing quicker response to critical resource shortages.
- Optimized database performance in **PostgreSQL** by redesigning table structures and writing efficient queries, improving reporting speed and reducing query time for large datasets by 25%.
- Enhanced forecasting capabilities using **NumPy and SciPy**, applying statistical modeling for scenario planning and improving the precision of resource allocation forecasts by 20%.
- Built a **Kafka-based** real-time data streaming pipeline, enabling hospital administrators to receive live updates on patient admissions, resource availability, and emergency alerts, thus improving responsiveness and operational coordination.
- Integrated **BigQuery** to automate data ingestion and querying, cutting manual data entry errors by 25% and significantly speeding up dashboard refresh times and ad hoc data requests.
- Ensured **HIPAA compliance** by implementing strict data encryption and access control mechanisms across all analytics processes, safeguarding patient privacy and adhering to regulatory standards while processing sensitive healthcare data.
- Managed project timelines and deliverables **using Jira**, coordinating cross-functional efforts between analytics, IT, and clinical teams, leading to a 15% improvement in delivery timelines and reporting accuracy.

EDUCATION

Master of Science in Computer science

University of North Texas

Aug 2023 - May 2025

Bachelor of Technology in computer science

Vel Tech University

Jun 2019 – Apr 2023

CERTIFICATIONS

- Data Analysis using Python
- AI-ML Data Analytics Internship
- AWS Academy Data Analytics
- AWS Academy Machine Learning